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Tutorial-6F

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Transfer Learning Strategies and Fine-tuning Best Practices.

- 1.) Pre-trained Model: ResNet-18. Trained on ImageNet. The final layer has 512 Input features and 1000 output classes.

Question.

1. Replace the 1000-class output Layer with a 2-class output layer. Calculate the number of trainable parameters in the new Layer, including bias.

$$\text{Parameter} = 512 \times 2 + 2$$

$$= 1024 + 2$$

$$= 1026$$

2Q:- If all ResNet-18 Layers except the new output Layer are frozen, identify which part of the network is trained.

Sol:- Only the newly added 2

class fully connected output Layer is trained.

35:- two models give the following validation accuracy. feature.

Extraction = 84% + Fine-tuning = 90.7.
which model performs better.

sol:- The Fine tuning model has the higher validation accuracy so it is better.

Q:- For the Fine-tuned model, 70 cat. images and 68 Dog images are correctly classified. Also, 5 cats are classified as Dogs and 7 Dogs are classified as cats. Calculate accuracy, precision, recall & f1-Score for the Dog class.

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	T Dog	P cat
+	68 (TP)	5 (FN)
-	7 (FN)	68 (TP)

Accuracy = $\frac{68 + 70}{68 + 7 + 5 + 70} = \frac{138}{150}$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$= \frac{68}{68+5} = \frac{68}{73}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$= \frac{68}{68+7} = \frac{68}{75}$$

F1-Score.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 2 \times \left(\frac{\frac{68}{73} \times \frac{68}{75}}{\frac{68}{73} + \frac{68}{75}} \right)$$

$$= 91.89\%$$

2.) Cats & Dogs Image Classification

$$\begin{aligned} 1) \quad 512 \times 2 &= 1024 \\ 1024 + 2 &= 1026 \end{aligned}$$

$$2) \quad 92 - 88 = 4\%$$

4% improvement.

$$3) \quad \alpha = 0.01.$$

$$\text{Accuracy} = 87\%$$

$$87 - 80 = 7\%$$

$\therefore 0.0001$ Learning rate gives higher validation accuracy

$$4) \quad \text{Total images} = 150$$

$$\text{Classified correct} = 138$$

$$= \frac{138}{150} \times 100 = 92\%$$

~~92%~~